Nazifa Mosharrat

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LinkedIn | GitHub | Portfolio

Professional Summary

Software Engineer with 5+ years of experience designing and developing scalable full-stack applications, backend systems, and business automation solutions across healthcare, logistics, finance, and enterprise domains. Strong hands on background in Python, Java, C#, REST APIs, databases, cloud deployment, and containerized applications. Specialized in Artificial Intelligence through a Master's in Artificial Intelligence at University of Pisa, where academic and research projects focused on Generative AI, Retrieval-Augmented Generation (RAG), LLM applications, NLP, and machine learning systems. Built practical AI solutions including intelligent assistants, automated reporting tools, and domain-specific RAG systems using LangChain, Weaviate, and open-source LLMs. Combines proven industry software engineering experience with current advanced AI expertise, bringing the ability to translate emerging AI capabilities into reliable real-world applications.

Professional Experience

Research Assistant

University of Pisa | Jan 2026– Present

- Developing an advanced Graph RAG system on a Linux-based server for the medical domain using Python, Weaviate, Qwen embeddings, BGE-M3, and BGE reranking, with systematic evaluation and debugging to improve retrieval accuracy, reliability, and system performance.
- Engineered an end-to-end Smart KPI automation and reporting system using Python, LangChain, LLaMA 3.2, Ollama, and Weaviate, integrating API-driven workflows, data processing, real-time query handling, and automated PDF/email report generation.
- Designed and validated multi-source data-processing and retrieval pipelines for document ingestion, embedding generation, vector search, reranking, contextual retrieval, and automated output generation.
- Led development of the MedPharm AI solution, integrating GPT-based models, RAG, and REST/API orchestration to automate medicine-waste tracking, data analysis, and supply-chain decision-support workflows.
- Implemented and evaluated NLP and machine-learning systems using BERT, LSTM, RNN, HMM, PyTorch, and transformer-based architectures, performing functional validation, comparative evaluation, and performance analysis across different models and configurations.
- Applied systematic testing, experimentation, debugging, error analysis, and data-driven validation to identify failures, verify expected system behaviour, and improve the performance, robustness, and reliability of AI/software pipelines.

Software Engineer (Full Stack & System Integration)

Xceleran | Remote (Pennsylvania, USA) | Jul 2021 – Sep 2024

- Designed and developed scalable enterprise applications for scheduling, charter services, POS, virtual payments, and franchisor financial management, supporting 4 franchisors and multiple franchisees across distributed business operations.
- Built and maintained REST API and third-party integration workflows for vehicle tracking, financial services, and enterprise applications, enabling reliable data exchange across external and internal systems.
- Implemented Single Sign-On (SSO) across multiple enterprise applications, improving centralized authentication, security, and maintainability.

- Developed an Pitch Deck Analyzer, integrating application logic, APIs, and automated document-processing workflows to streamline analysis and reporting.
- Collaborated with engineering and business teams to troubleshoot production systems, investigate integration issues, improve backend performance, and enhance application scalability and reliability.

Jr. Programmer (Full Stack Developer)

Neosysworld | Chattogram, Bangladesh | Nov 2020 – Aug 2021

- Developed travel/ticket management, courier logistics, inventory, and budgeting systems.
- Built "NSW Ticket" chatbot system (rule-based) for automated query response.
- Worked across database design, web development, and software deployment cycles.

Jr. Programmer – IT Department

Chattogram Maa-O-Shishu-O General Hospital | Oct 2019 – Oct 2020

- Developed ERP modules for **Pharmacy Management, HR Management, and COVID-19 Test Booth Automation**, supporting clinical and administrative workflow automation.
- Designed database structures and automated workflows for the COVID-19 Test Booth system, supporting reliable data capture, processing, storage, and end-to-end system operation.
- Assisted with **software implementation, functional validation, system-level testing, and troubleshooting** of clinical management systems, gaining practical exposure to hardware-software interaction, failure identification, and overall system reliability.

Education

- Master's Degree in Artificial Intelligence | University of Pisa, Italy | 2023– 2026
- BSc in Computer Science and Engineering (Professional Bachelor) | East Delta University, Bangladesh | 2019–2022 | GPA: 3.58/4.00
- Diploma in Computer Technology | Chittagong Mohila Polytechnic Institute, Bangladesh | 2013–2018 | CGPA: 3.84/4.00

Skills

- **Programming Languages:** Python, C#, Java
- **Machine Learning & AI Frameworks:** PyTorch, TensorFlow, Scikit-Learn, Transformers, Hugging Face, LangChain, GPT-3,Medgemma,Qwen, Ollama 3.2, Stanza, BERT, LLaMA, GANs, CycleGen, StyleGAN, OpenCV, RAG (Retrieval-Augmented Generation)
- **Generative AI & LLM Technologies:** Fine-tuning LLMs, Prompt Engineering, RLHF (Reinforcement Learning from Human Feedback), Model Quantization (GGUF), LoRA, API Integration for AI Systems
- **Vector Databases & Knowledge Retrieval:** Weaviate (for RAG), FAISS
- **Data Science & Analytics Tools:** Pandas, NumPy
- **Backend & API Development:** REST APIs, FastAPI, .NET Core, ASP.NET, Spring Boot
- **Frontend & Web Technologies (GUI):** AngularJS, Thymeleaf, HTML5, Razor, Blazor
- **Reporting Tools:** SAP Crystal Reports, FastReport .NET, JasperReports
- **DevOps & MLOps:** Docker, AWS (EC2), GitHub, Bitbucket, Jira, CI/CD, Model Deployment & Optimization
- **Databases:** PostgreSQL, MySQL, Microsoft SQL Server
- **Cloud & Scalable AI Systems:** AWS, API-based Orchestration, Microservices
- **Integration & Business APIs:** OpenAI API, Google API, Payeezy, Quickbook, ZeoTab, Tua Pay, Wisetack

Certifications and Training

- **Generative AI with Diffusion Models** | NVIDIA Deep Learning Institute | Technical – Advanced
  Hands-on training on text-to-image generation using diffusion models and PyTorch, U-Net architecture, Forward/reverse diffusion, Denoising, CLIP for prompt-based image generation, Model optimization using GELU, GroupNorm, and sinusoidal embeddings.
- **Building RAG Agents with LLMs** | Self-Paced Course | Technical – Advanced
  Hands-on training on deploying RAG-based agentic systems using large language models, implementing dialog management and internal reasoning, using embeddings for semantic retrieval and guardrailing, designing pipelines with LangChain, Gradio, and LangServe, orchestrating document interactions, configuring vector stores for content retrieval, and evaluating modular RAG agents without fine-tuning.
- **Generative AI** | NVIDIA Deep Learning Institute | Technical – Beginner
  Overview of Generative AI concepts and applications using neural networks.
- **Statistics for Data Science** | Great Learning | Issued Jan 2021
  Fundamentals of statistical analysis, probability, hypothesis testing, regression, and data interpretation for data science.
- **Machine Learning with Python** | Great Learning | Issued Dec 2020
  Introduction to supervised and unsupervised learning, model evaluation, feature engineering, and implementation in Python.
- **Advanced Game Development** | ICT Division (Government of Bangladesh) | Issued Sep 2016
  Game architecture, logic development, gameplay programming, and performance optimization techniques.
- **Game Graphics Design** | ICT Division (Government of Bangladesh) | Issued Sep 2016
  Visual design principles, 2D/3D asset creation, character modelling, UI/UX for games, and creative tools usage.